

Arrays

tipos

bool

int

long

float

double

char

string

...

bool 1 byte

int 4 bytes

long 8 bytes

float 4 bytes

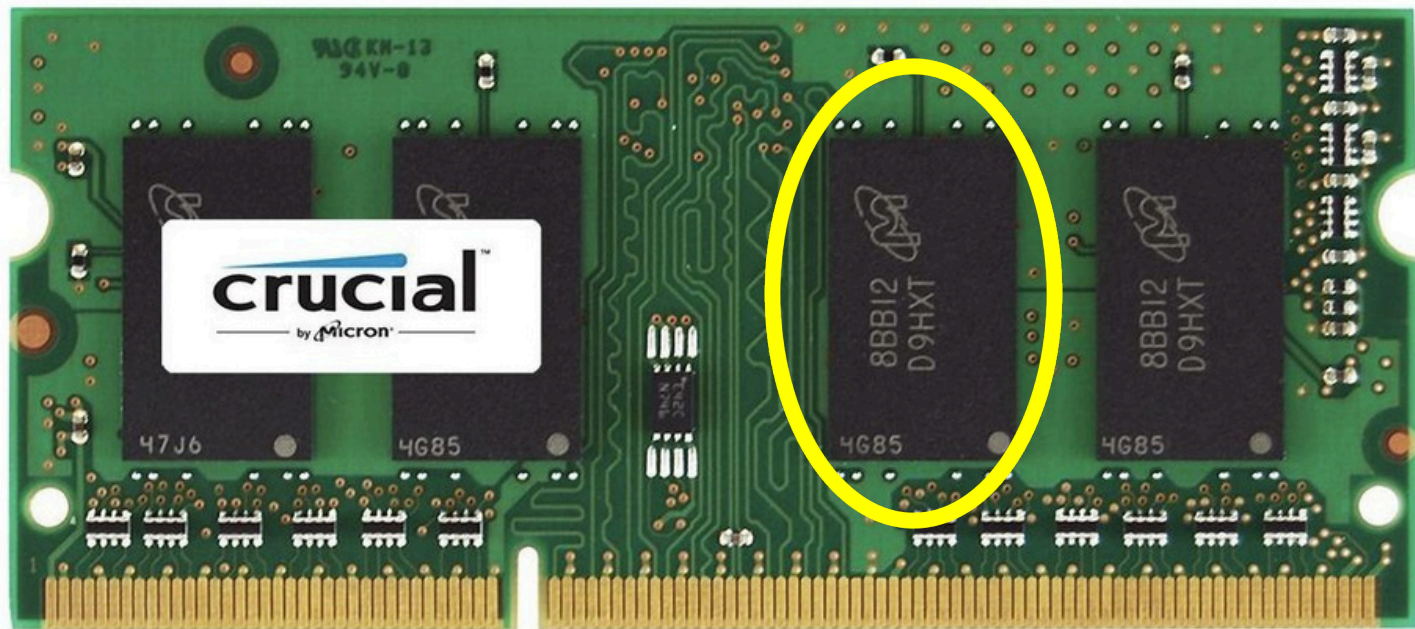
double 8 bytes

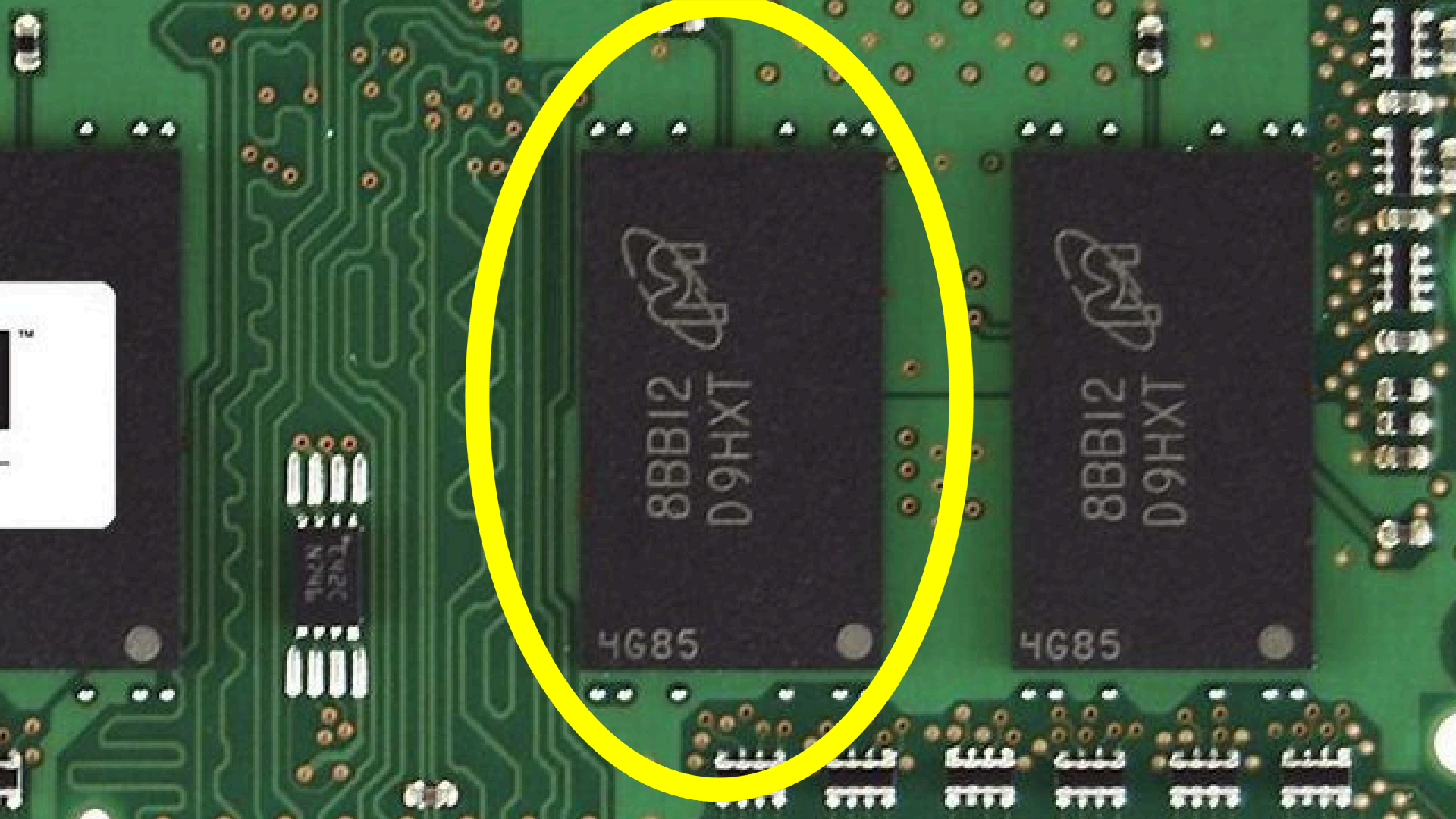
char 1 byte

string ? bytes

...

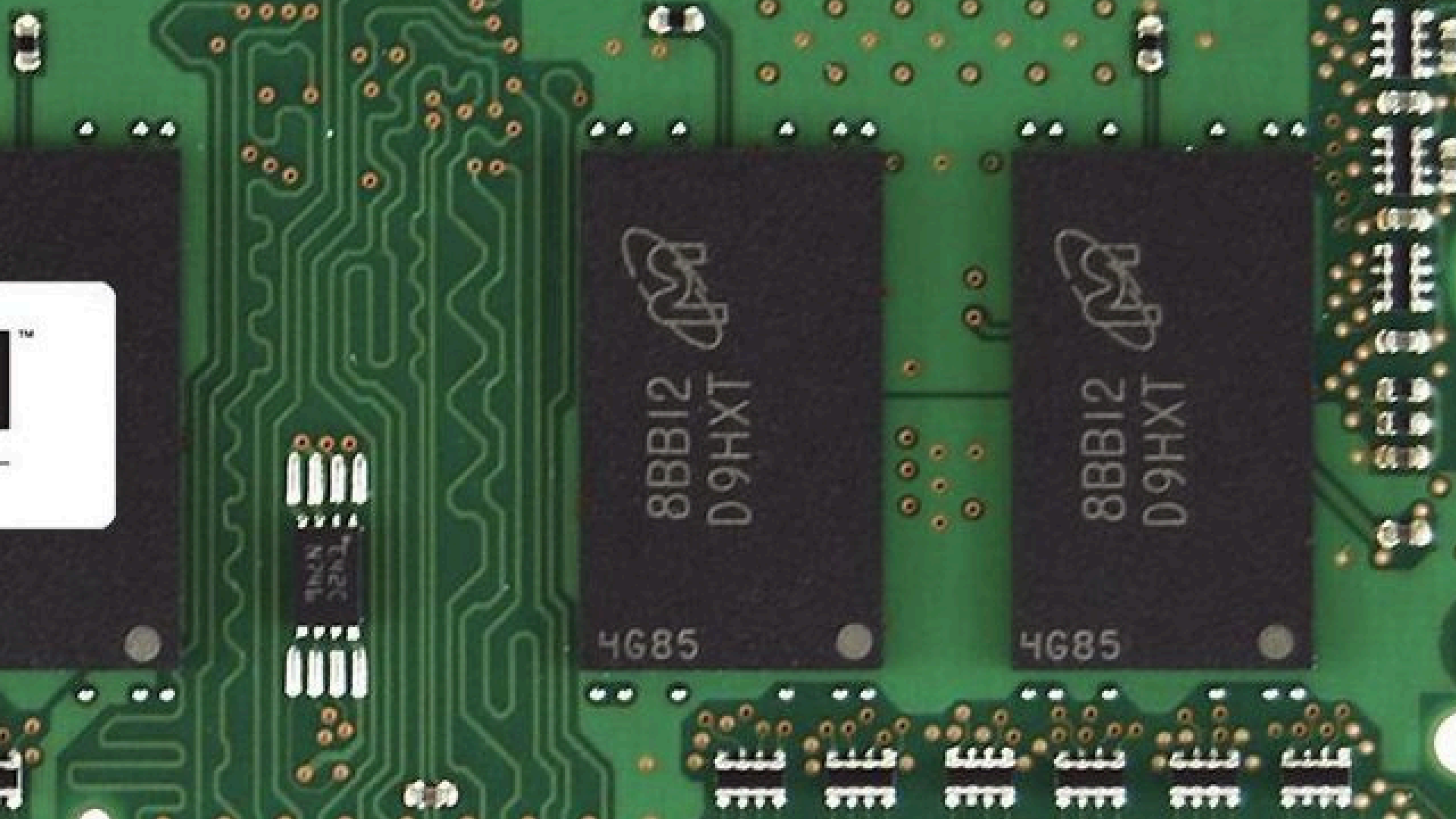






8BB12
D9HXT

4G85



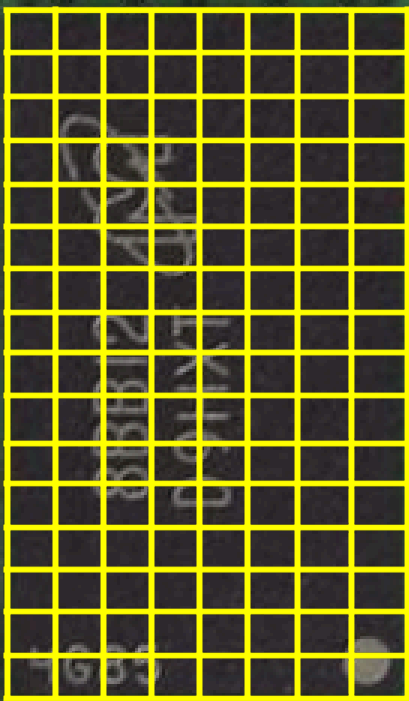
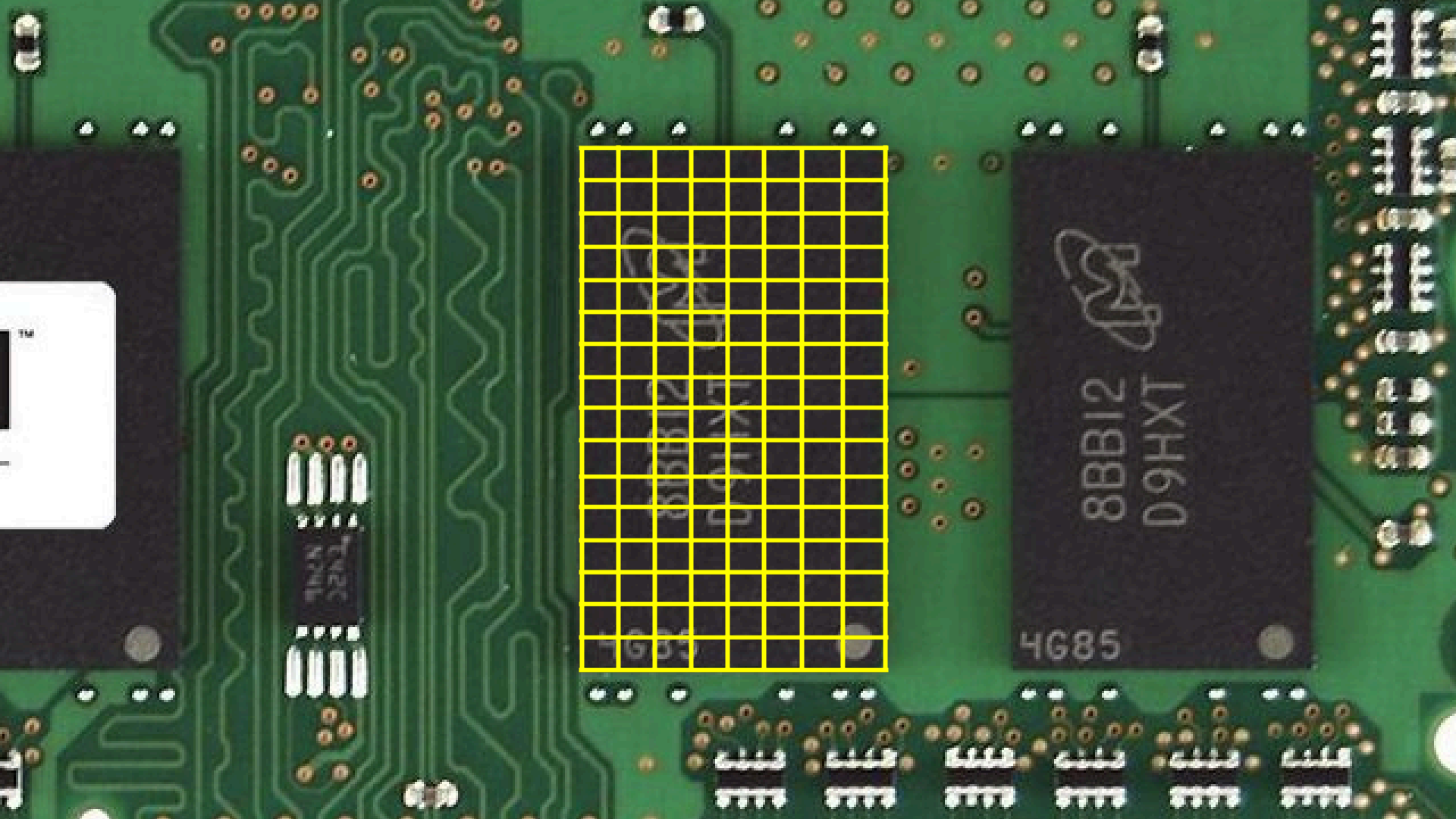
8BB12
D9HXT

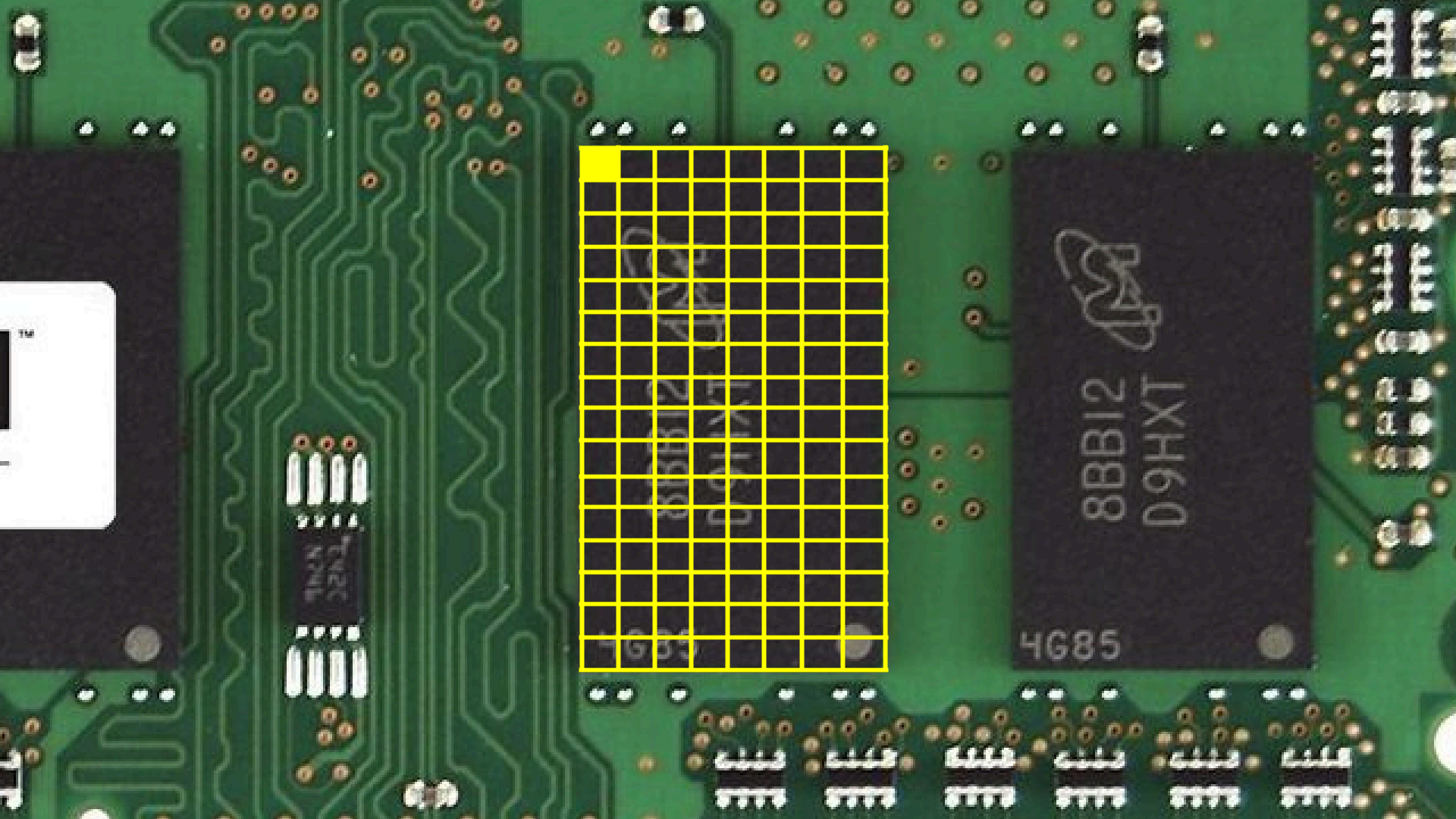
4G85

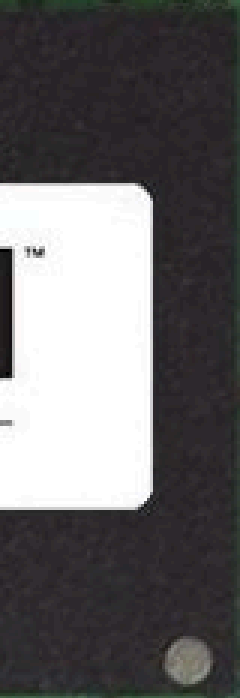
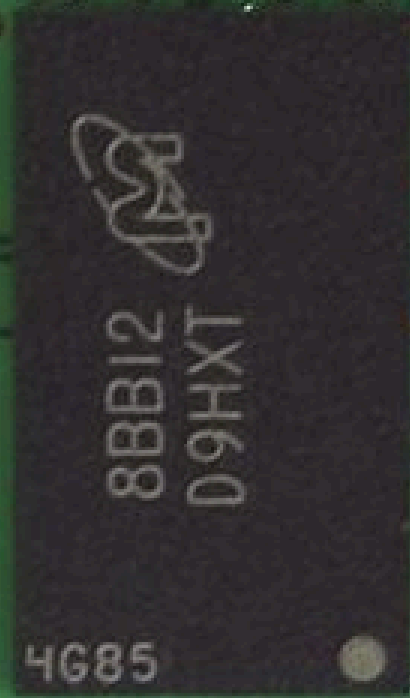
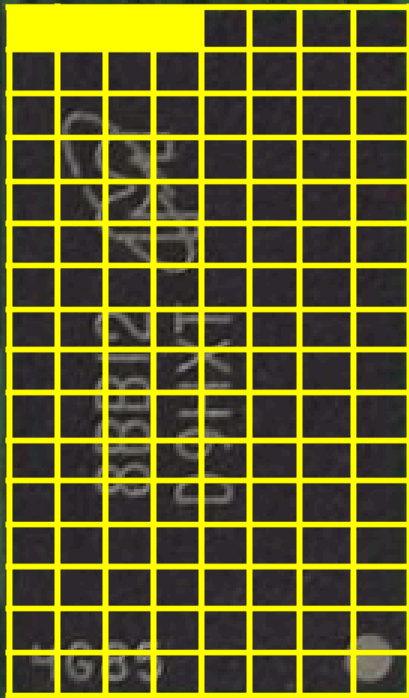
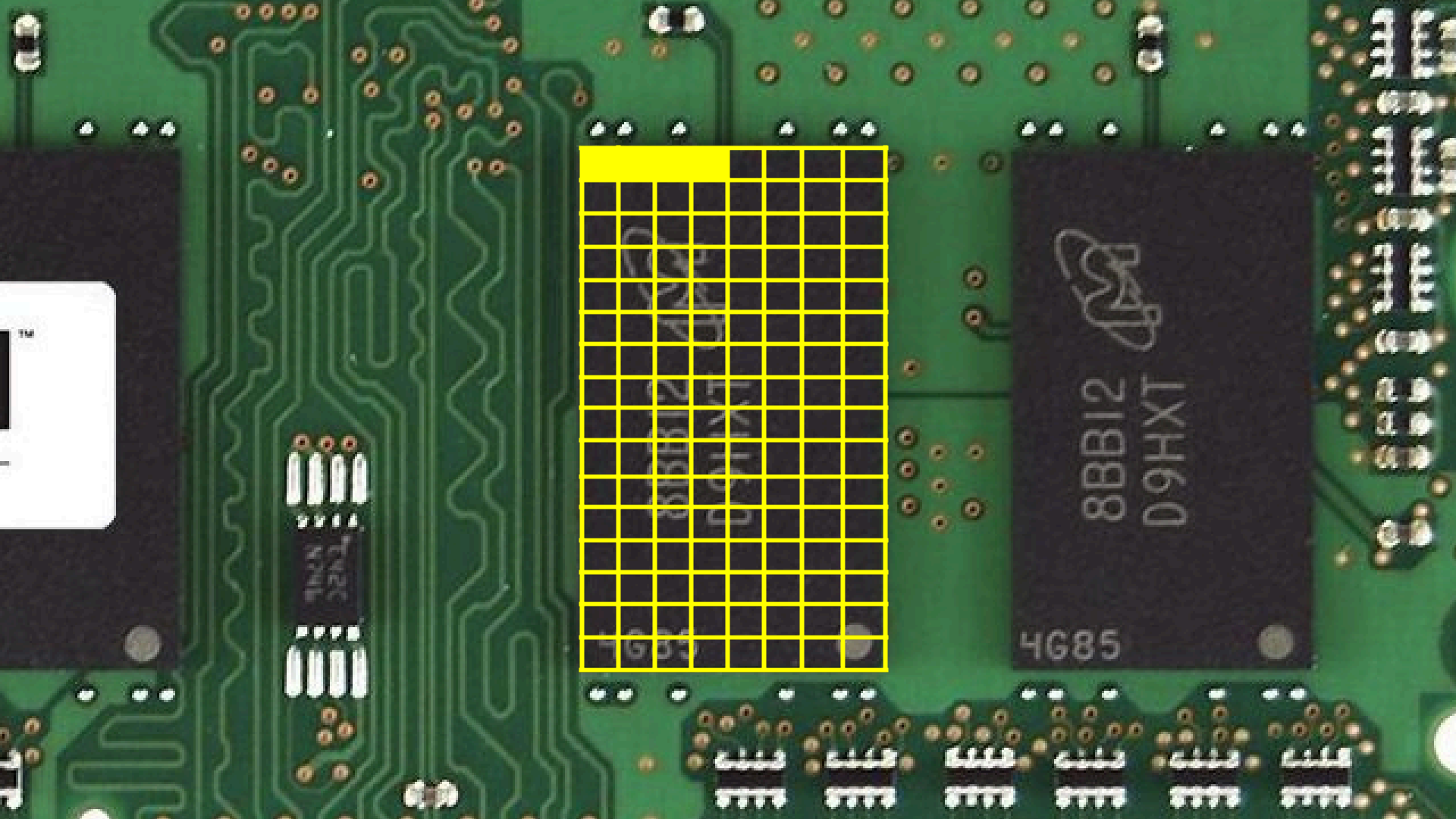
8BB12
D9HXT

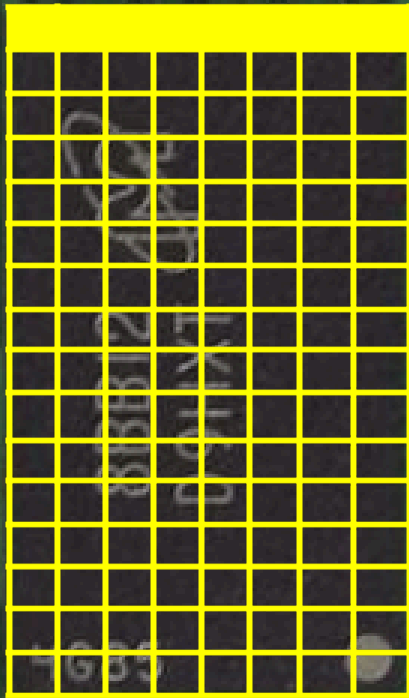
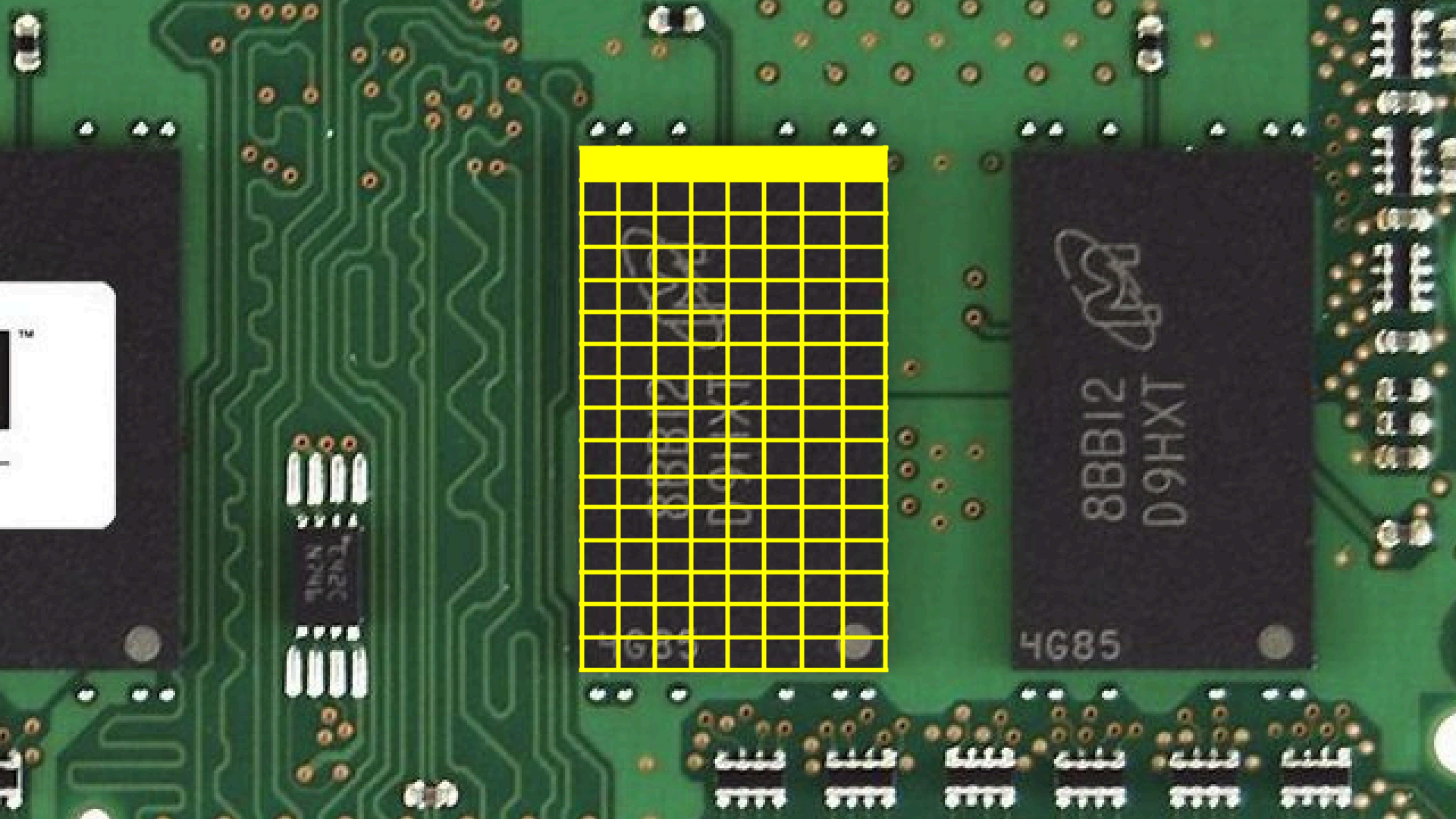
4G85

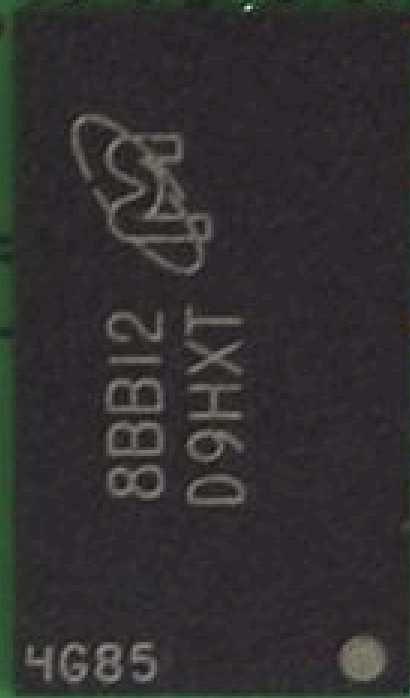
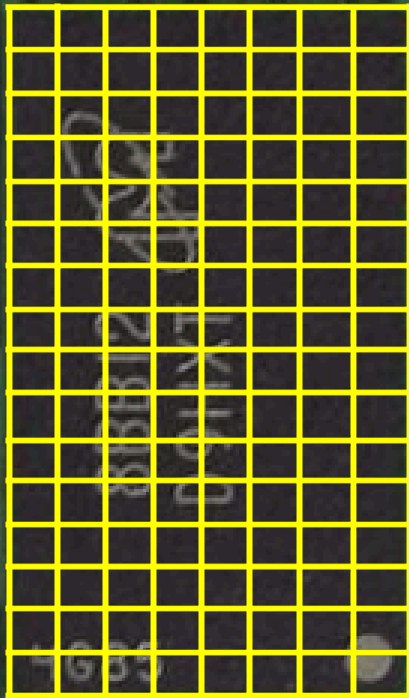
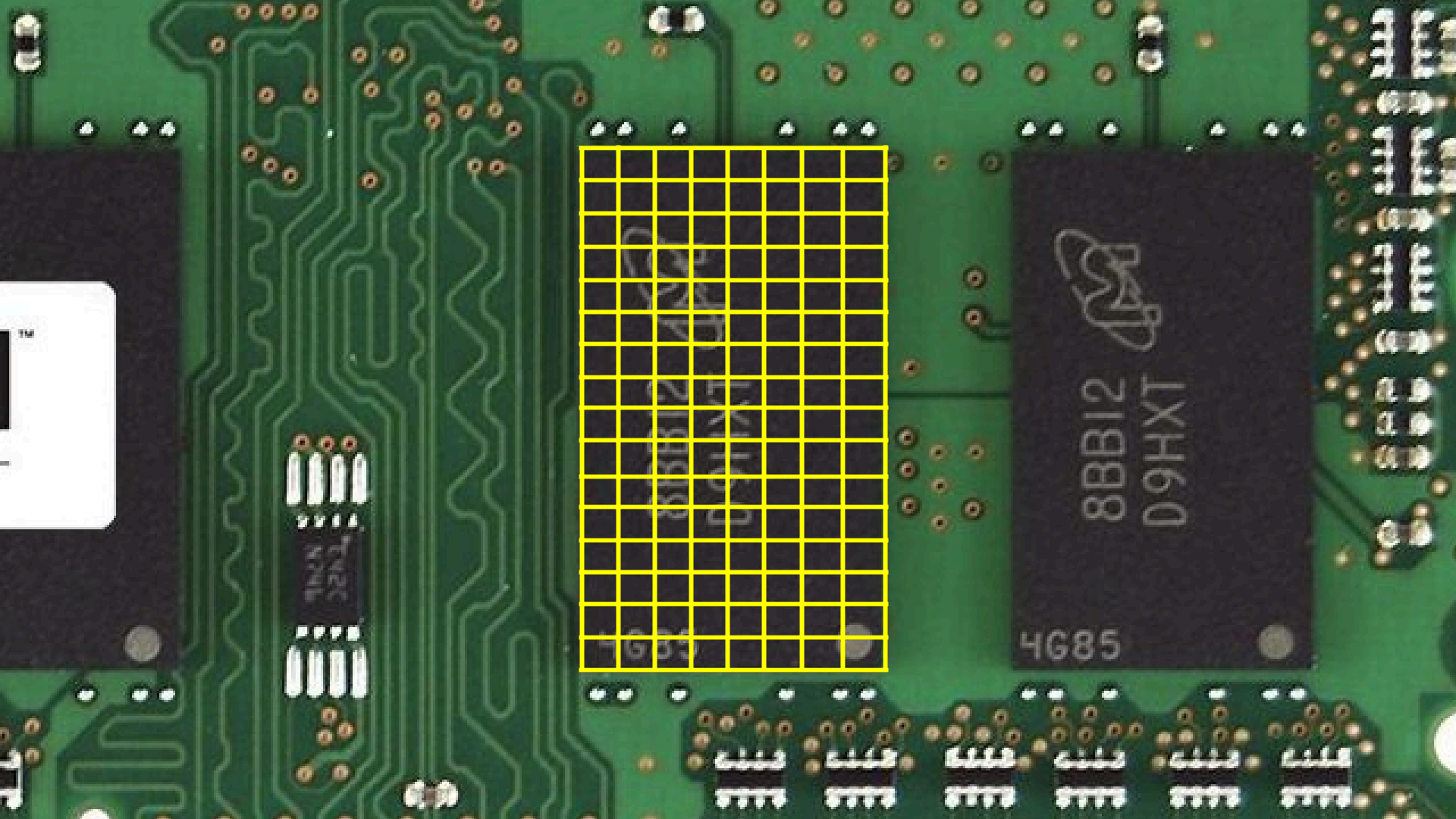
™

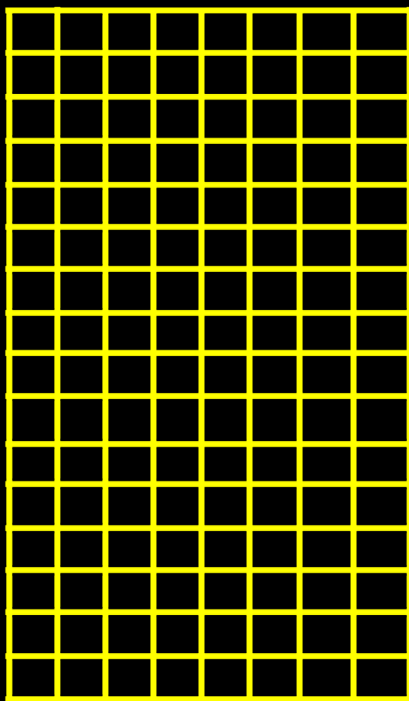


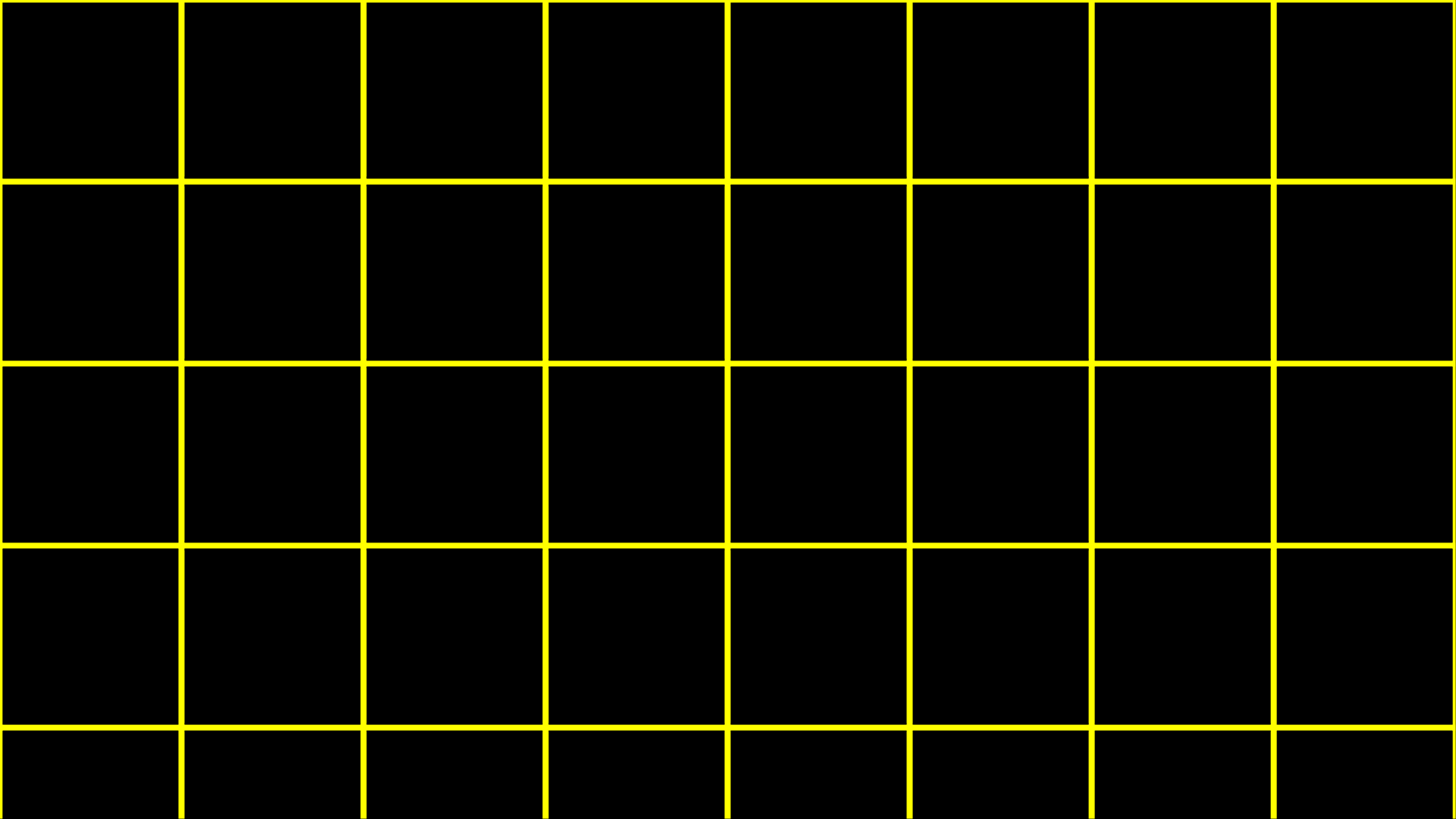








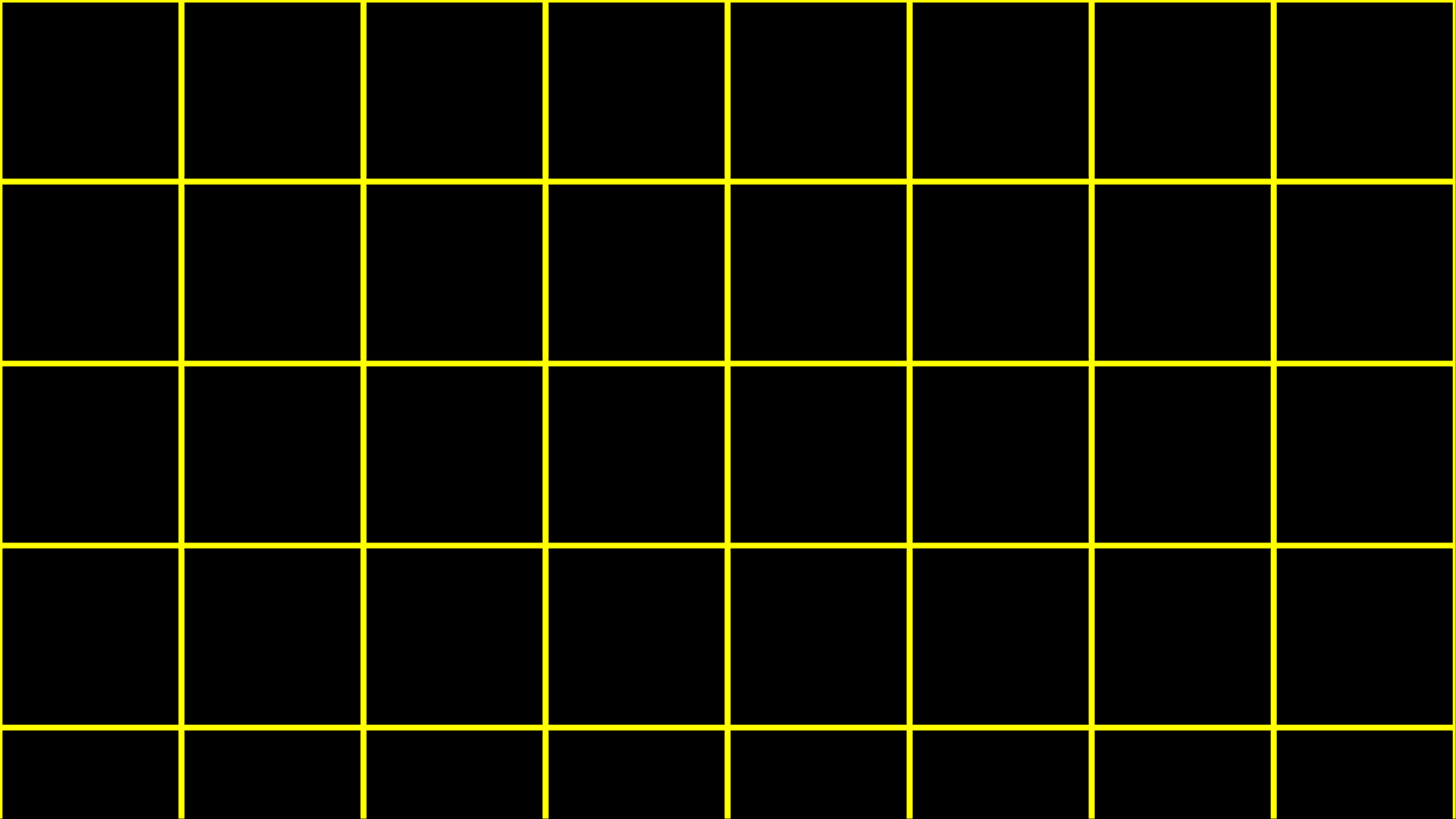




```
int ponto1 = 79;
```

```
int ponto2 = 73;
```

```
int ponto3 = 33;
```

79

ponto1

79

ponto1

73

ponto2

79

ponto1

73

ponto2

33

ponto3

000000000000000000000000000000001001111 ponto1				000000000000000000000000000000001001001 ponto2			
00000000000000000000000000000000100001 ponto3							

```
int ponto1 = 79;
```

```
int ponto2 = 73;
```

```
int ponto3 = 33;
```

arrays

```
int pontos[3];
```



```
int pontos[3];
```

```
pontos[0] = 79;
```

```
pontos[1] = 73;
```

```
pontos[2] = 33;
```

72

pontos[0]

73

pontos[1]

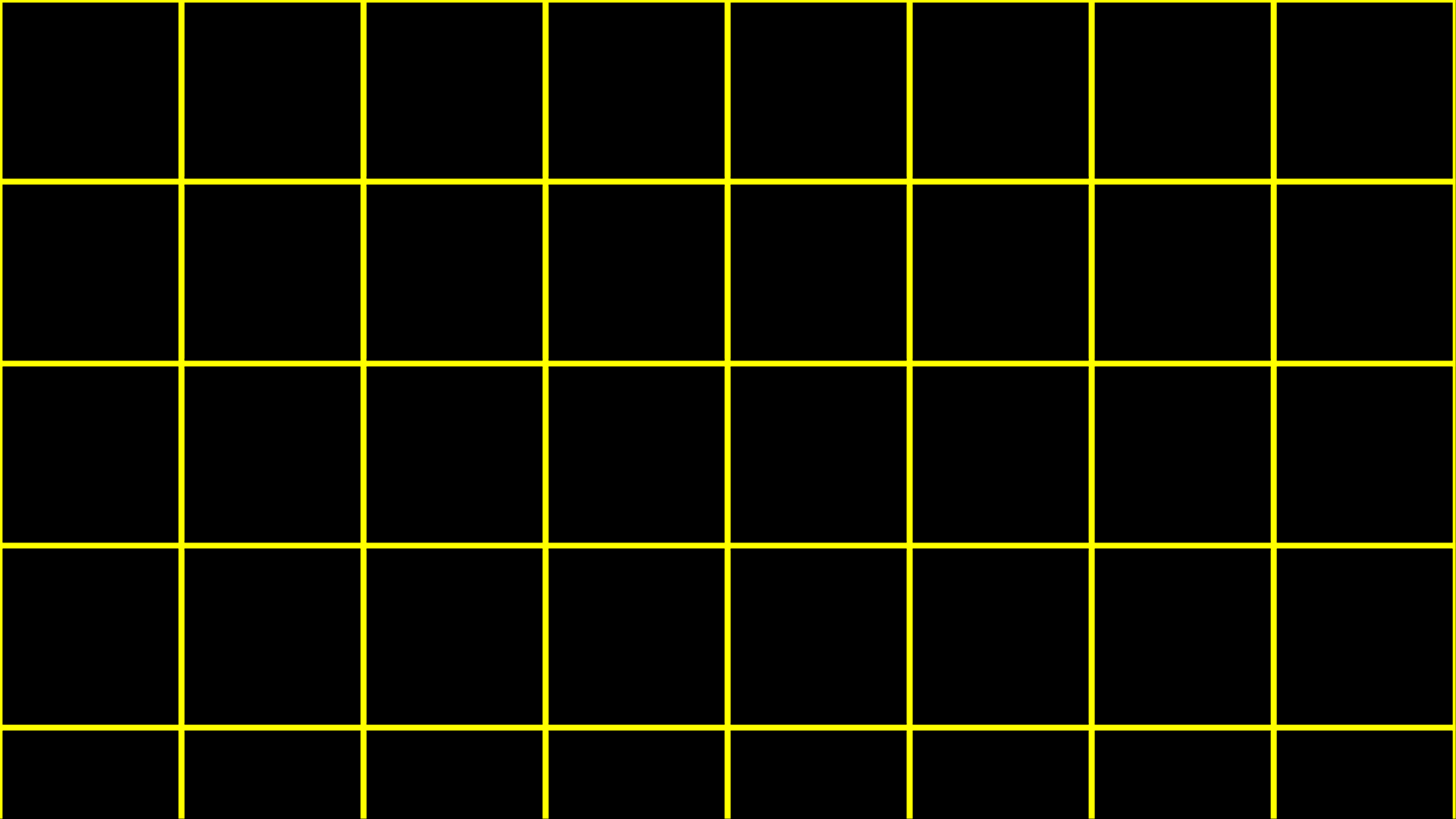
33

pontos[2]

```
char c1 = '0';
```

```
char c2 = 'I';
```

```
char c3 = '!';
```



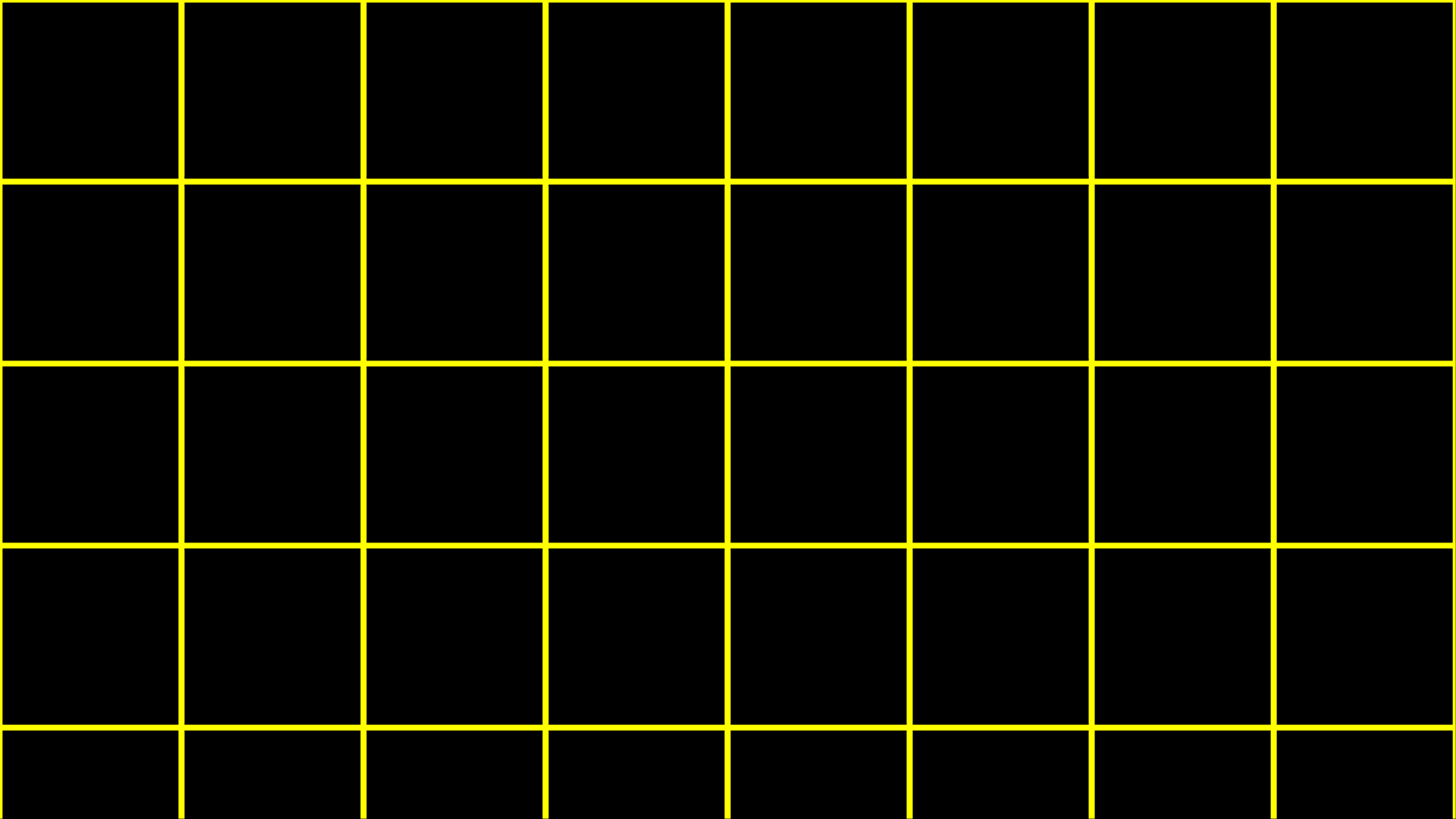
0 c1	I c2	! c3					

79	73	33					

01001111 c1	01001001 c2	00100001 c3					

string


```
string s = "OI!";
```



0	I	!					
	s						

0 s[0]	I s[1]	! s[2]					

0 s[0]	I s[1]	! s[2]	00000000 s[3]				

0 s[0]	I s[1]	! s[2]	∅ s[3]				

0 s[0]	I s[1]	! s[2]	\0 s[3]				

79	73	33	0 s[3]				

0

I

!

\0

s

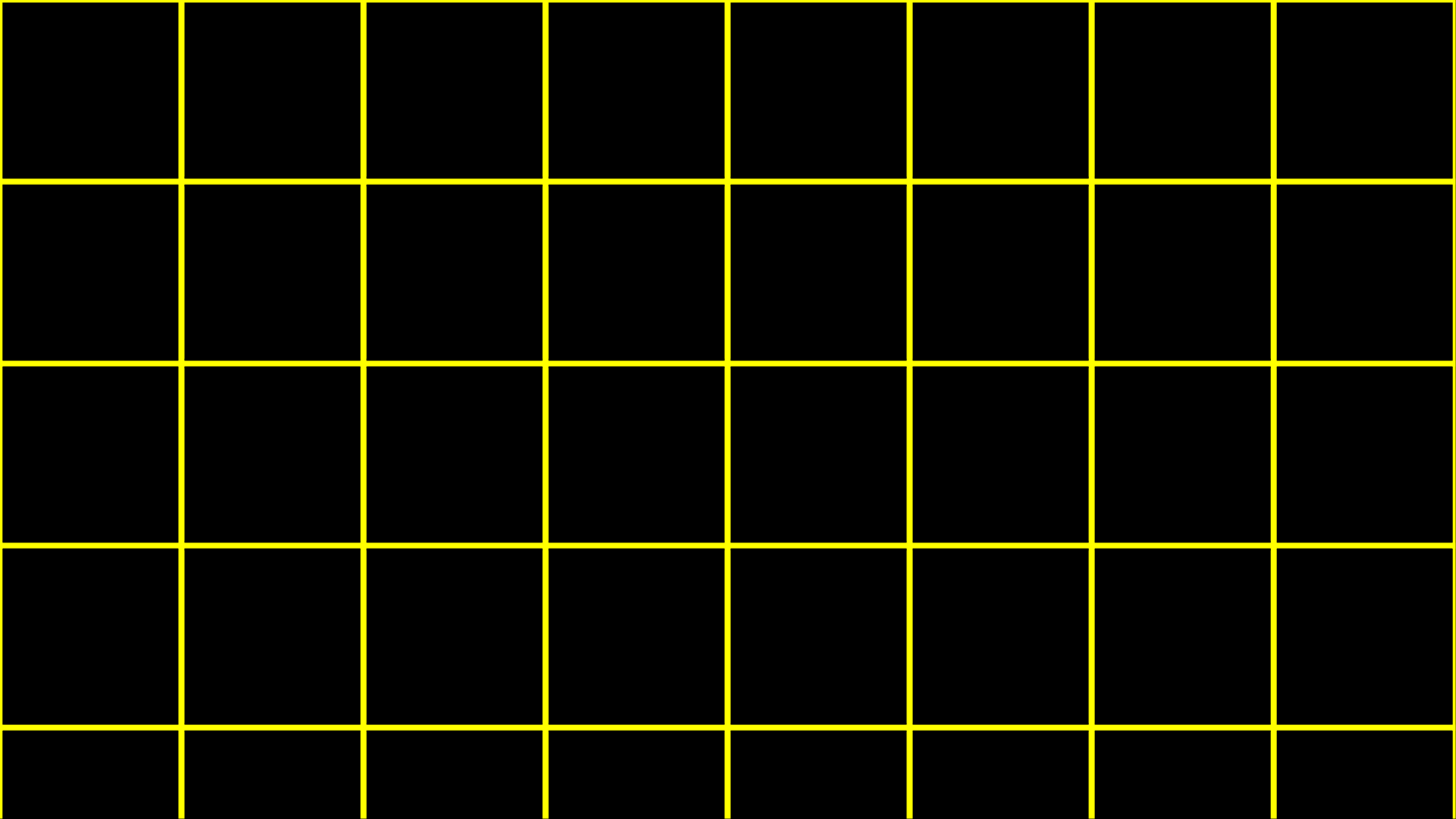
NUL

0	<u>NUL</u>	16	<u>DLE</u>	32	<u>SP</u>	48	0	64	@	80	P	96	`	112	p
1	<u>SOH</u>	17	<u>DC1</u>	33	!	49	1	65	A	81	Q	97	a	113	q
2	<u>STX</u>	18	<u>DC2</u>	34	"	50	2	66	B	82	R	98	b	114	r
3	<u>ETX</u>	19	<u>DC3</u>	35	#	51	3	67	C	83	S	99	c	115	s
4	<u>EOT</u>	20	<u>DC4</u>	36	\$	52	4	68	D	84	T	100	d	116	t
5	<u>ENQ</u>	21	<u>NAK</u>	37	%	53	5	69	E	85	U	101	e	117	u
6	<u>ACK</u>	22	<u>SYN</u>	38	&	54	6	70	F	86	V	102	f	118	v
7	<u>BEL</u>	23	<u>ETB</u>	39	'	55	7	71	G	87	W	103	g	119	w
8	<u>BS</u>	24	<u>CAN</u>	40	(	56	8	72	H	88	X	104	h	120	x
9	<u>HT</u>	25	<u>EM</u>	41	)	57	9	73	I	89	Y	105	i	121	y
10	<u>LF</u>	26	<u>SUB</u>	42	*	58	:	74	J	90	Z	106	j	122	z
11	<u>VT</u>	27	<u>ESC</u>	43	+	59	;	75	K	91	[	107	k	123	{
12	<u>FF</u>	28	<u>FS</u>	44	,	60	<	76	L	92	\	108	l	124	
13	<u>CR</u>	29	<u>GS</u>	45	-	61	=	77	M	93	]	109	m	125	}
14	<u>SO</u>	30	<u>RS</u>	46	.	62	>	78	N	94	^	110	n	126	~
15	<u>SI</u>	31	<u>US</u>	47	/	63	?	79	O	95	_	111	o	127	<u>DEL</u>

0	<u>NUL</u>	16	<u>DLE</u>	32	<u>SP</u>	48	0	64	@	80	P	96	`	112	p
1	<u>SOH</u>	17	<u>DC1</u>	33	!	49	1	65	A	81	Q	97	a	113	q
2	<u>STX</u>	18	<u>DC2</u>	34	"	50	2	66	B	82	R	98	b	114	r
3	<u>ETX</u>	19	<u>DC3</u>	35	#	51	3	67	C	83	S	99	c	115	s
4	<u>EOT</u>	20	<u>DC4</u>	36	\$	52	4	68	D	84	T	100	d	116	t
5	<u>ENQ</u>	21	<u>NAK</u>	37	%	53	5	69	E	85	U	101	e	117	u
6	<u>ACK</u>	22	<u>SYN</u>	38	&	54	6	70	F	86	V	102	f	118	v
7	<u>BEL</u>	23	<u>ETB</u>	39	'	55	7	71	G	87	W	103	g	119	w
8	<u>BS</u>	24	<u>CAN</u>	40	(	56	8	72	H	88	X	104	h	120	x
9	<u>HT</u>	25	<u>EM</u>	41	)	57	9	73	I	89	Y	105	i	121	y
10	<u>LF</u>	26	<u>SUB</u>	42	*	58	:	74	J	90	Z	106	j	122	z
11	<u>VT</u>	27	<u>ESC</u>	43	+	59	;	75	K	91	[	107	k	123	{
12	<u>FF</u>	28	<u>FS</u>	44	,	60	<	76	L	92	\	108	l	124	
13	<u>CR</u>	29	<u>GS</u>	45	-	61	=	77	M	93	]	109	m	125	}
14	<u>SO</u>	30	<u>RS</u>	46	.	62	>	78	N	94	^	110	n	126	~
15	<u>SI</u>	31	<u>US</u>	47	/	63	?	79	O	95	_	111	o	127	<u>DEL</u>

```
string s = "OI!";
```

```
string t = "ATE!";
```



0 I ! \0							
s							

0 I ! \0				A T E !			
s				t			
\0							

0 s[0]	I s[1]	! s[2]	\0 s[3]	A t[0]	T t[1]	E t[2]	! t[3]
\0 t[4]							

```
string palavras[2];
```

```
palavras[0] = "OI!";
```

```
palavras[1] = "ATE!";
```

O I ! \0

A T E !

palavras[0]

palavras[1]

\0

O	I	!	\0	A	T	E	!
palavras[0][1]	palavras[0][1]	palavras[0][2]	palavras[0][3]	palavras[1][0]	palavras[1][1]	palavras[1][2]	palavras[1][3]
\0							
palavras[1][4]							

string

string.h

manual.cs50.io/#string.h

strlen

ctype.h

manual.cs50.io/#ctype.h

0	<u>NUL</u>	16	<u>DLE</u>	32	<u>SP</u>	48	0	64	@	80	P	96	`	112	p
1	<u>SOH</u>	17	<u>DC1</u>	33	!	49	1	65	A	81	Q	97	a	113	q
2	<u>STX</u>	18	<u>DC2</u>	34	"	50	2	66	B	82	R	98	b	114	r
3	<u>ETX</u>	19	<u>DC3</u>	35	#	51	3	67	C	83	S	99	c	115	s
4	<u>EOT</u>	20	<u>DC4</u>	36	\$	52	4	68	D	84	T	100	d	116	t
5	<u>ENQ</u>	21	<u>NAK</u>	37	%	53	5	69	E	85	U	101	e	117	u
6	<u>ACK</u>	22	<u>SYN</u>	38	&	54	6	70	F	86	V	102	f	118	v
7	<u>BEL</u>	23	<u>ETB</u>	39	'	55	7	71	G	87	W	103	g	119	w
8	<u>BS</u>	24	<u>CAN</u>	40	(	56	8	72	H	88	X	104	h	120	x
9	<u>HT</u>	25	<u>EM</u>	41	)	57	9	73	I	89	Y	105	i	121	y
10	<u>LF</u>	26	<u>SUB</u>	42	*	58	:	74	J	90	Z	106	j	122	z
11	<u>VT</u>	27	<u>ESC</u>	43	+	59	;	75	K	91	[	107	k	123	{
12	<u>FF</u>	28	<u>FS</u>	44	,	60	<	76	L	92	\	108	l	124	
13	<u>CR</u>	29	<u>GS</u>	45	-	61	=	77	M	93	]	109	m	125	}
14	<u>SO</u>	30	<u>RS</u>	46	.	62	>	78	N	94	^	110	n	126	~
15	<u>SI</u>	31	<u>US</u>	47	/	63	?	79	O	95	_	111	o	127	<u>DEL</u>

0	<u>NUL</u>	16	<u>DLE</u>	32	<u>SP</u>	48	0	64	@	80	P	96	`	112	p
1	<u>SOH</u>	17	<u>DC1</u>	33	!	49	1	65	A	81	Q	97	a	113	q
2	<u>STX</u>	18	<u>DC2</u>	34	"	50	2	66	B	82	R	98	b	114	r
3	<u>ETX</u>	19	<u>DC3</u>	35	#	51	3	67	C	83	S	99	c	115	s
4	<u>EOT</u>	20	<u>DC4</u>	36	\$	52	4	68	D	84	T	100	d	116	t
5	<u>ENQ</u>	21	<u>NAK</u>	37	%	53	5	69	E	85	U	101	e	117	u
6	<u>ACK</u>	22	<u>SYN</u>	38	&	54	6	70	F	86	V	102	f	118	v
7	<u>BEL</u>	23	<u>ETB</u>	39	'	55	7	71	G	87	W	103	g	119	w
8	<u>BS</u>	24	<u>CAN</u>	40	(	56	8	72	H	88	X	104	h	120	x
9	<u>HT</u>	25	<u>EM</u>	41	)	57	9	73	I	89	Y	105	i	121	y
10	<u>LF</u>	26	<u>SUB</u>	42	*	58	:	74	J	90	Z	106	j	122	z
11	<u>VT</u>	27	<u>ESC</u>	43	+	59	;	75	K	91	[	107	k	123	{
12	<u>FF</u>	28	<u>FS</u>	44	,	60	<	76	L	92	\	108	l	124	
13	<u>CR</u>	29	<u>GS</u>	45	-	61	=	77	M	93	]	109	m	125	}
14	<u>SO</u>	30	<u>RS</u>	46	.	62	>	78	N	94	^	110	n	126	~
15	<u>SI</u>	31	<u>US</u>	47	/	63	?	79	O	95	_	111	o	127	<u>DEL</u>

main

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    ...
```

```
}
```

```
#include <stdio.h>
```

```
int main(void)  
{  
    ...  
}
```

```
#include <stdio.h>
```

```
int main(int argc, string argv[])  
{  
    ...  
}
```


exit status

An unknown error occurred

Error code: 1132

[Report Problem](#)

OK

404

This is not the
web page you
are looking for.



```
#include <stdio.h>
```

```
int main(int argc, string argv[])  
{  
    ...  
}
```

```
#include <stdio.h>
```

```
int main(int argc, string argv[])  
{  
    ...  
}
```

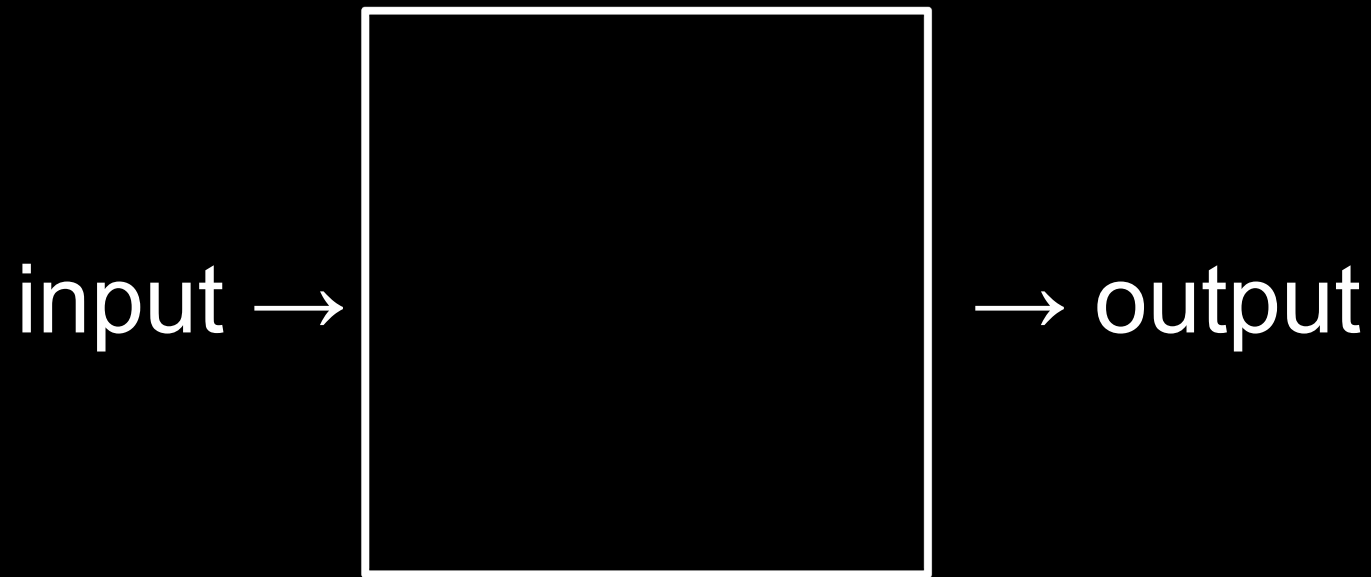
```
#include <stdio.h>
```

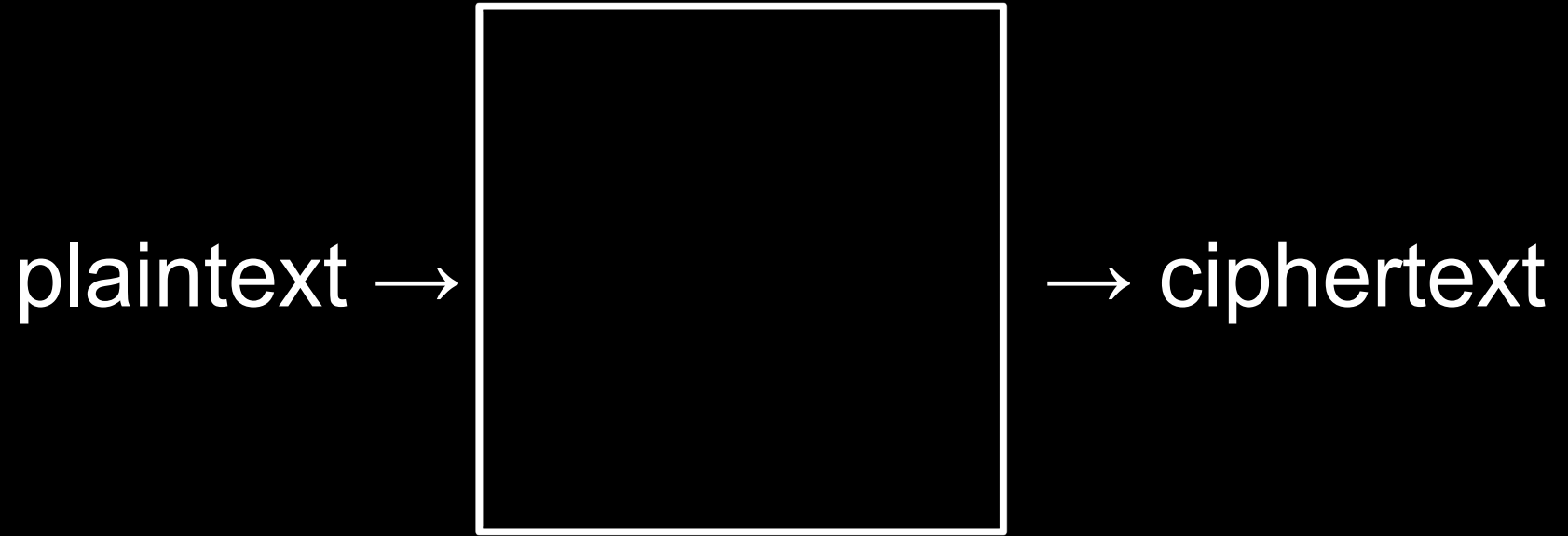
```
int main(void)  
{  
    ...  
}
```

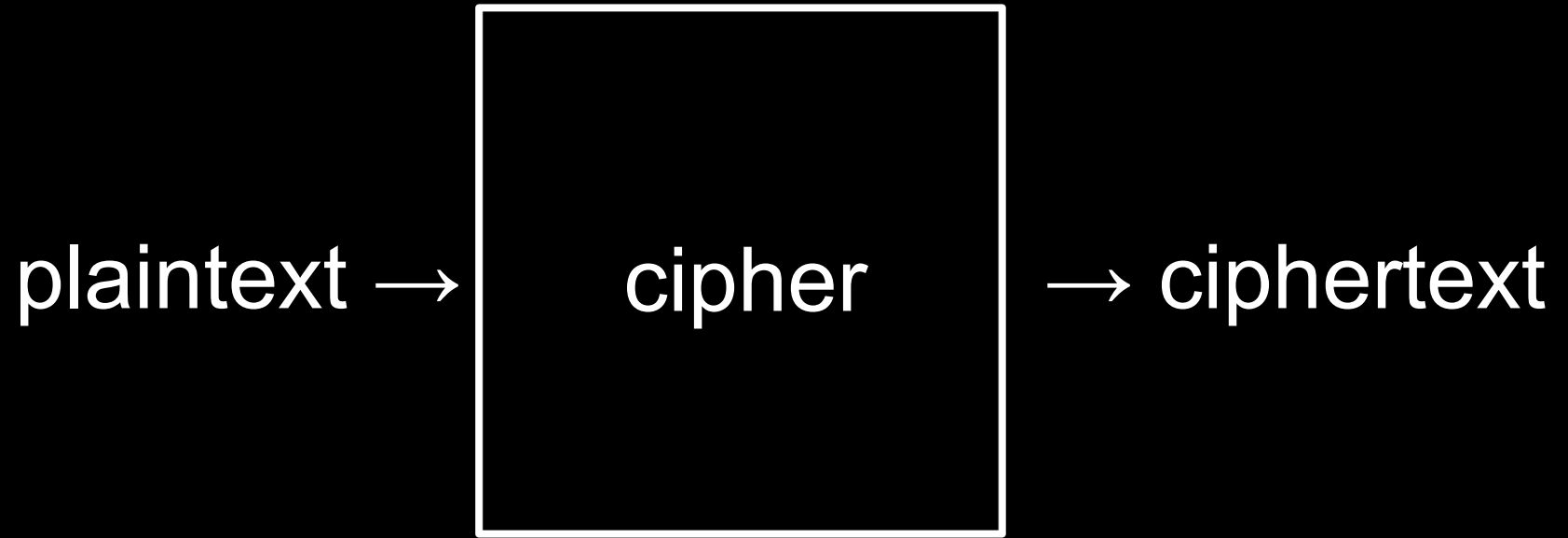
echo \$?

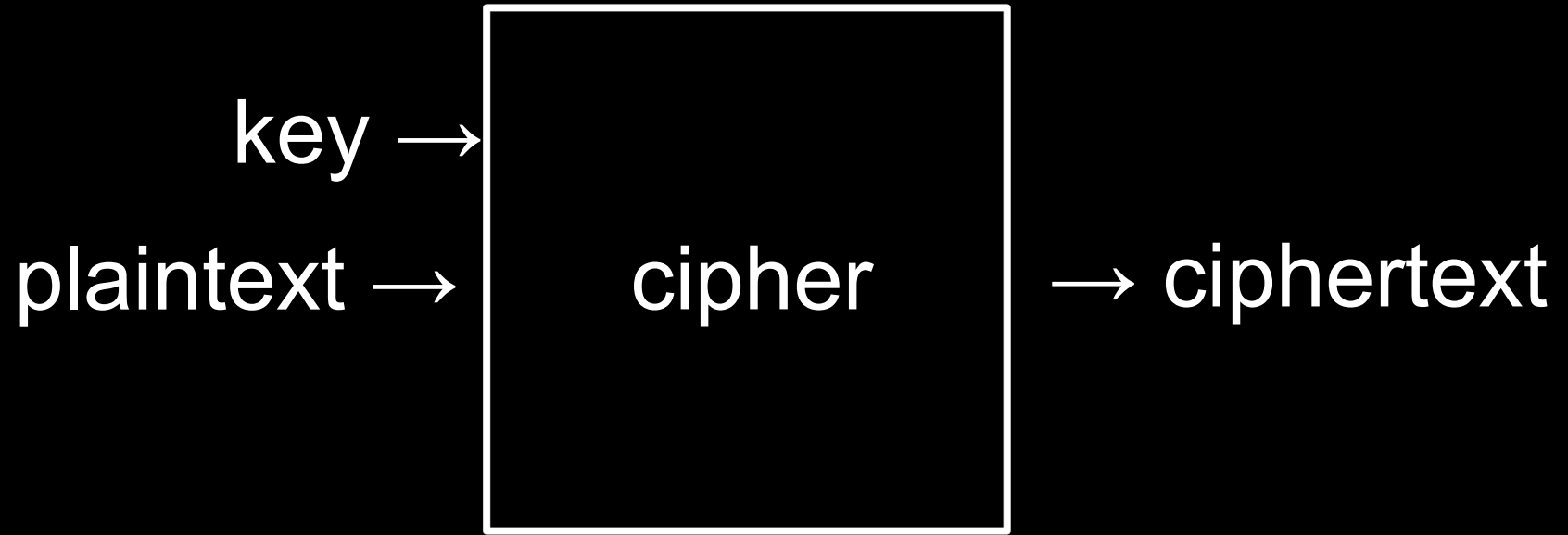
criptografia

encryption



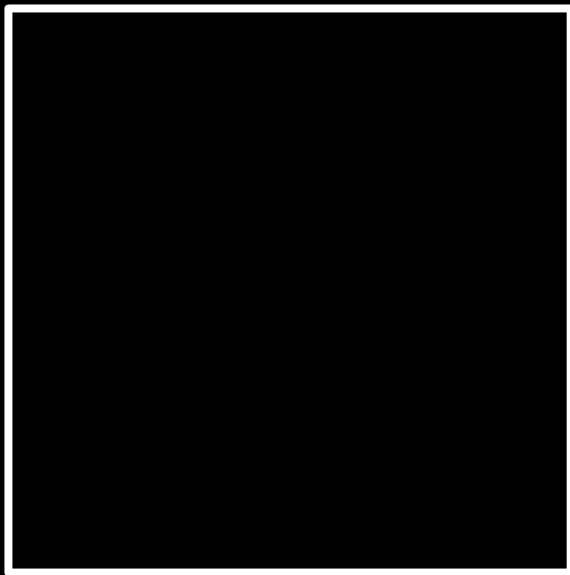


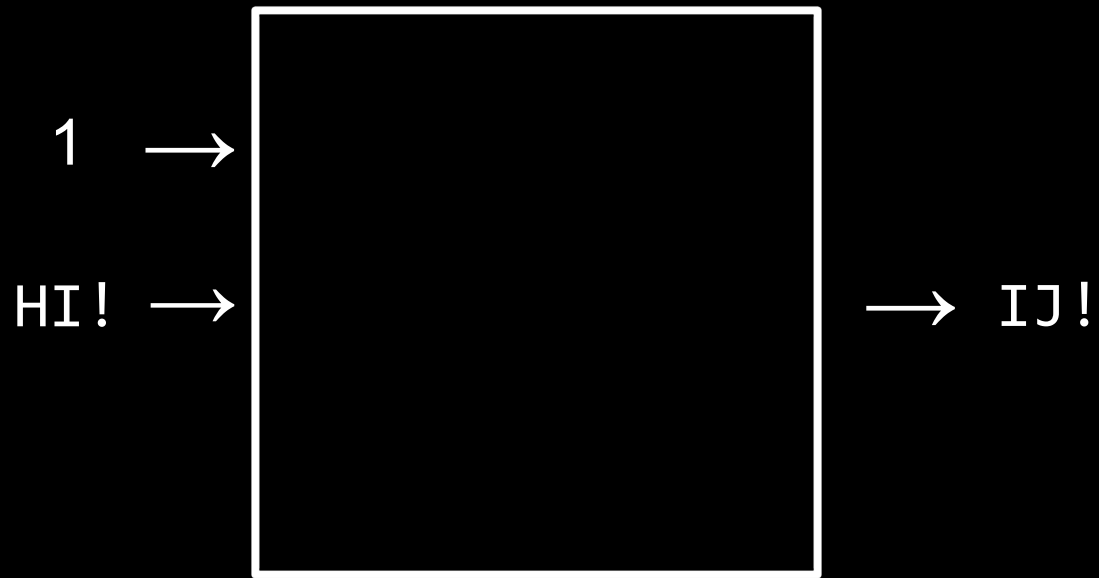


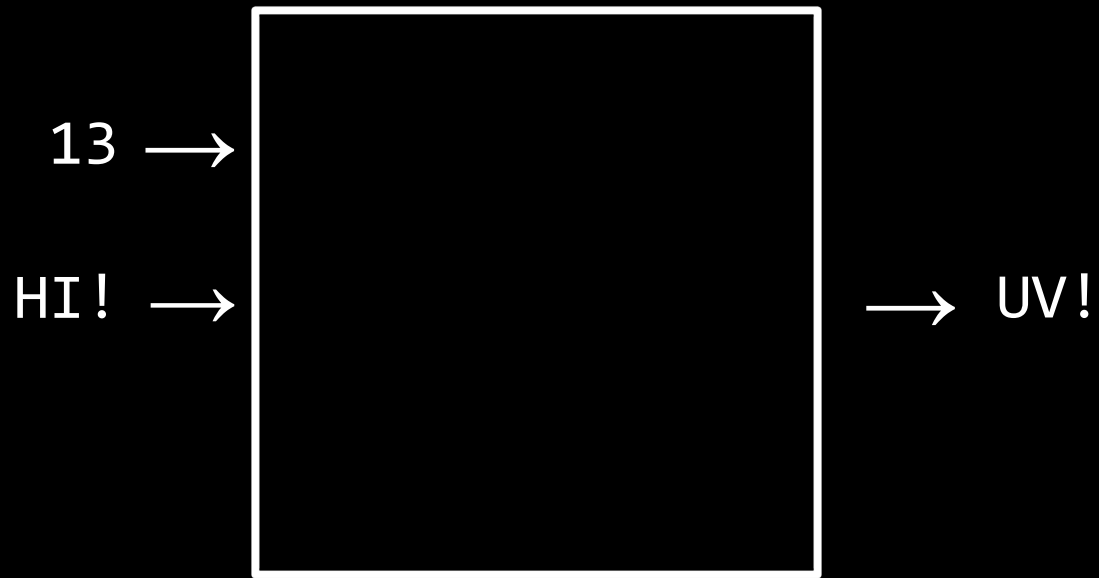


1 →

HI! →

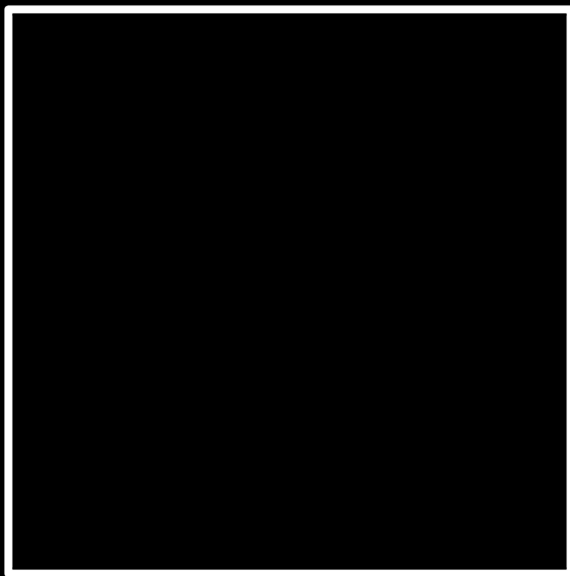






13 →

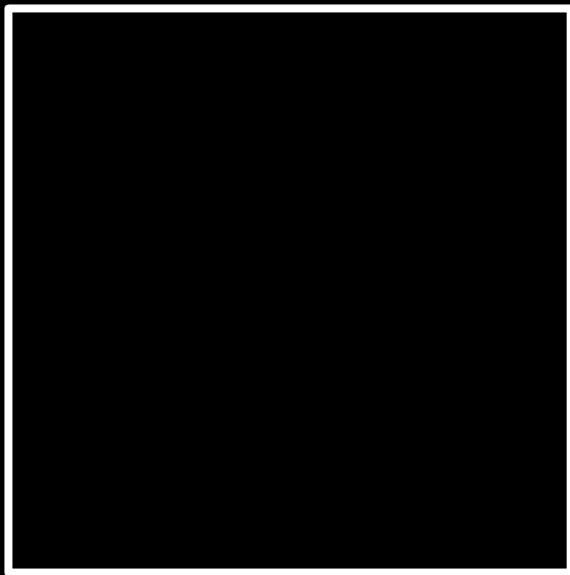
I LOVE YOU →



→ V YBIR LBH

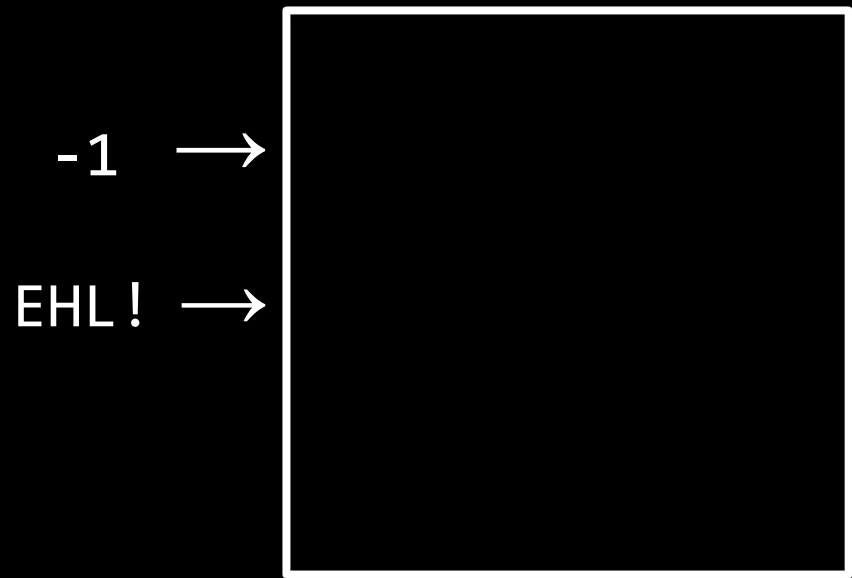
26 →

I LOVE YOU →





decryption



G J N !

F J N !

F I N !

F I M !

FIM!

